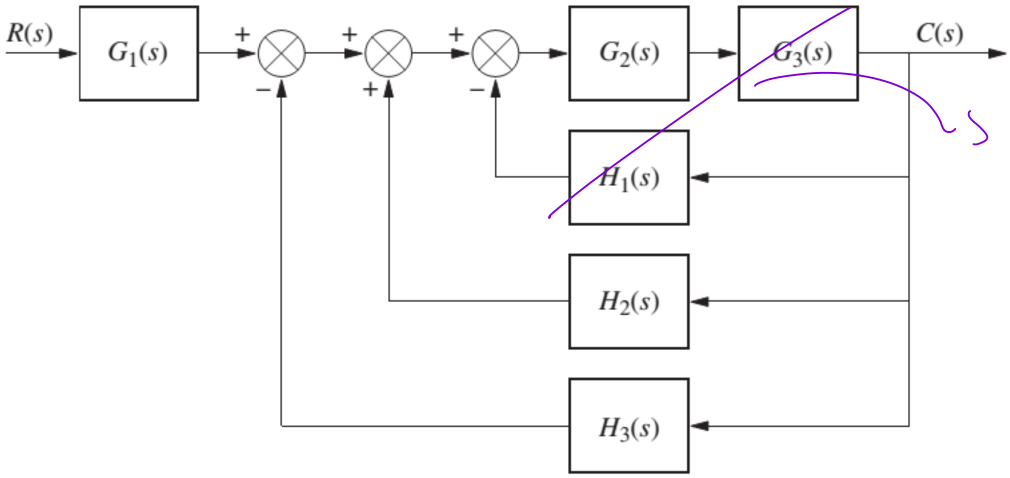
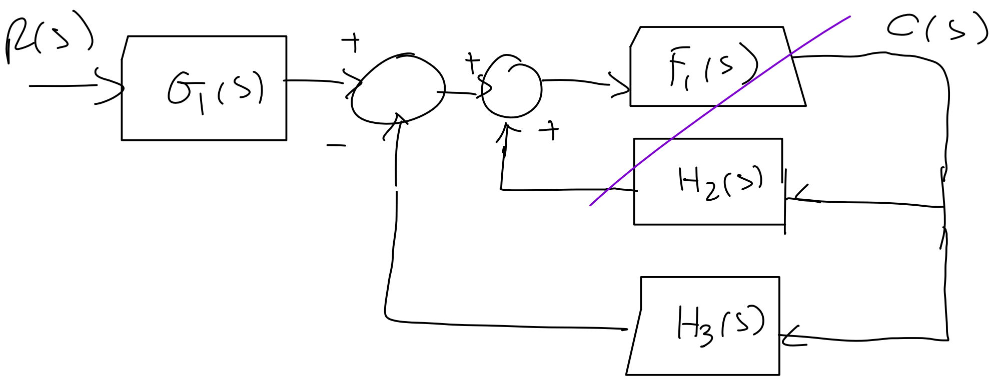
Assignment 5 (ELEC 341 L5_Model_DCMotor)

Problem 1:

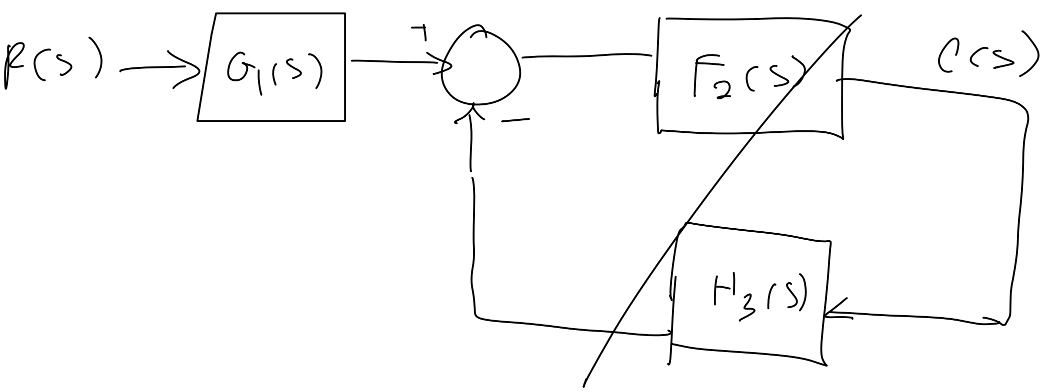
Reduce the block diagram shown in figure to a single transfer function:



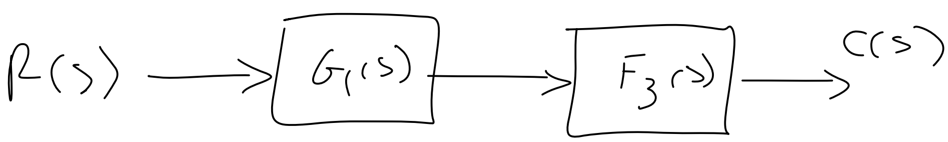
$$\frac{G_2(s)G_3(s)}{1+G_2(s)G_3(s)H_1(s)} = F_1(s)$$



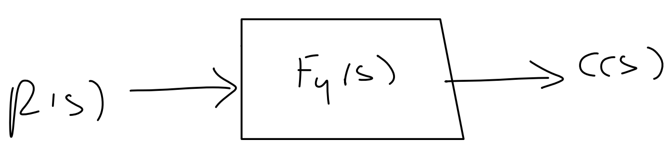
$$\frac{F_1(s)}{1-F_1(s)H_2(s)} = F_2(s).$$



$$\frac{F_2(s)}{1+F_2(s)H_3(s)} = F_3(s)$$



$$F_4(s) = G_1(s)F_3(s)$$



$$\frac{C(s)}{R(s)} = F_4(s).$$

$$F_1(s) = \frac{G_2(s)G_3(s)}{1 + G_2(s)G_3(s)H_1(s)}$$

$$F_2(s) = \frac{F_1(s)}{1 - F_1(s)H_2(s)} = \frac{\dfrac{G_2(s)G_3(s)}{1+G_2(s)G_3(s)H_1(s)}}{1 - \dfrac{G_2(s)G_3(s)H_2(s)}{1+G_2(s)G_3(s)H_1(s)}}$$

$$F_2(s) = \frac{G_2(s)G_3(s)}{1+G_2(s)G_3(s)H_1(s) - G_2(s)G_3(s)H_2(s)}$$

$$F_3(s) = \frac{F_2(s)}{1+F_2(s)H_3(s)} = \frac{\dfrac{G_2(s)G_3(s)}{1+G_2(s)G_3(s)H_1(s) - G_2(s)G_3(s)H_2(s)}}{1 + \dfrac{G_2(s)G_3(s)H_3(s)}{1+G_2(s)G_3(s)H_1(s) - G_2(s)G_3(s)H_2(s)}}$$

$$F_3(s) = \frac{G_2(s)G_3(s)}{1+G_2(s)G_3(s)H_1(s) - G_2(s)G_3(s)H_2(s) + G_2(s)G_3(s)H_3(s)}$$

$$F_4(s) = \frac{C(s)}{R(s)} = G_1(s)F_3(s) = \boxed{\frac{G_1(s)G_2(s)G_3(s)}{1+G_2(s)G_3(s)H_1(s) - G_2(s)G_3(s)H_2(s) + G_2(s)G_3(s)H_3(s)}}$$

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Problem 2:

$v_0' = 0.$

Find the linearize form of the following system using the equilibrium point of $v_0 = 30$. That is, re-write the ODE in terms of perturbation variables:

(v_0, F_0)

$$m\dot{v}(t) + K_f \cdot v(t) + K_a . v^2(t) = F_m(t)$$

Use the following numerical values for the constant parameters in your calculations:
$m = 1000$; $K_f = 0.1$; $K_a = 1$

$$f(v', v, F) = m v'(t) + K_f v(t) + K_a v^2(t) - F(t) = 0.$$

$$f(v', v, F) = 0 = \frac{\partial f}{\partial v'}(v' - v_0') + \frac{\partial f}{\partial v}(v - v_0) + \frac{\partial f}{\partial F}(F - F_0) = 0.$$
$$+ f(v_0', v_0, F_0)$$

find F_0. $\cancel{m v_0'}^{0} + K_f v_0 + \cancel{K_a}^{1} v_0^2 - F_0 = 0.$ so $(v_0, F_0) = (30, 903)$

$\cancel{(0.1)(30)}^{3} + 900 - F_0 = 0.$

$$\boxed{F_0 = 903}$$

$$f(v', v, F) = \frac{\partial f}{\partial v'}\Big|_{op} \delta v' + \frac{\partial f}{\partial v}\Big|_{op} \delta v + \frac{\partial f}{\partial F}\Big|_{op} \delta F + f(v_0', v_0, F_0)$$

where $\delta v' = v - v_0'$

$\delta v = v - v_0$

$\delta F = F - F_0.$

$$\frac{\partial f}{\partial v'}\Big|_{op} = m, \quad \frac{\partial f}{\partial v}\Big|_{op} = 2K_a v_0 + K_f, \quad \frac{\partial f}{\partial F}\Big|_{op} = -1.$$

$$\Rightarrow f(v', v, F) = m\,\delta v' + (2\cancel{K_a}^{1} v_0 + K_f)\,\delta v - \delta F + \cancel{f(v_0', v_0, F_0)}^{0}$$

$$1000\,\delta v' + 60.1\,\delta v - \delta F = 0$$

$$\boxed{1000\,\delta v' + 60.1\,\delta v = \delta F}$$